```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Load dataset
df = pd.read_csv("Dataset .csv")

# Show first 5 rows
df.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Currency	Has Table booking	Has Online delivery	Is delivering now	Switch to order menu	Price range	Aggregate rating	Rating color	Rating text	Votes
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botswana Pula(P)	Yes	No	No	No	3	4.8	Dark Green	Excellent	314
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Botswana Pula(P)	Yes	No	No	No	3	4.5	Dark Green	Excellent	591
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Botswana Pula(P)	Yes	No	No	No	4	4.4	Green	Very Good	270
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.056475	14.585318	Japanese, Sushi	...	Botswana Pula(P)	No	No	No	No	4	4.9	Dark Green	Excellent	365
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.057508	14.584450	Japanese, Korean	...	Botswana Pula(P)	Yes	No	No	No	4	4.8	Dark Green	Excellent	229

5 rows × 21 columns

```
# Number of rows and columns
df.shape
```

```
(9551, 21)
```

```
# Display all column names
df.columns
```

```
Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
       'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
       'Average Cost for two', 'Currency', 'Has Table booking',
       'Has Online delivery', 'Is delivering now', 'Switch to order menu',
       'Price range', 'Aggregate rating', 'Rating color', 'Rating text',
       'Votes'],
      dtype='object')
```

```
# Dataset information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   Restaurant ID         9551 non-null   int64  
 1   Restaurant Name       9551 non-null   object 
 2   Country Code          9551 non-null   int64  
 3   City                  9551 non-null   object 
 4   Address               9551 non-null   object 
 5   Locality              9551 non-null   object 
 6   Locality Verbose      9551 non-null   object 
 7   Longitude             9551 non-null   float64
 8   Latitude              9551 non-null   float64
 9   Cuisines              9542 non-null   object 
 10  Average Cost for two  9551 non-null   int64  
 11  Currency              9551 non-null   object 
 12  Has Table booking     9551 non-null   object 
 13  Has Online delivery   9551 non-null   object 
 14  Is delivering now     9551 non-null   object 
 15  Switch to order menu  9551 non-null   object 
 16  Price range           9551 non-null   int64  
 17  Aggregate rating      9551 non-null   float64
 18  Rating color          9551 non-null   object 
 19  Rating text           9551 non-null   object 
 20  Votes                 9551 non-null   int64  
dtypes: float64(3), int64(5), object(13)
memory usage: 1.5+ MB
```

```
# Missing values
df.isnull().sum()
```

	0
Restaurant ID	0
Restaurant Name	0
Country Code	0
City	0
Address	0
Locality	0
Locality Verbose	0
Longitude	0
Latitude	0
Cuisines	9
Average Cost for two	0
Currency	0
Has Table booking	0
Has Online delivery	0
Is delivering now	0
Switch to order menu	0
Price range	0
Aggregate rating	0
Rating color	0
Rating text	0
Votes	0

dtype: int64

```
# Statistical summary
df.describe()
```

	Restaurant ID	Country Code	Longitude	Latitude	Average Cost for two	Price range	Aggregate rating	Votes
count	9.551000e+03	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000
mean	9.051128e+06	18.365616	64.126574	25.854381	1199.210763	1.804837	2.666370	156.909748
std	8.791521e+06	56.750546	41.467058	11.007935	16121.183073	0.905609	1.516378	430.169145
min	5.300000e+01	1.000000	-157.948486	-41.330428	0.000000	1.000000	0.000000	0.000000
25%	3.019625e+05	1.000000	77.081343	28.478713	250.000000	1.000000	2.500000	5.000000
50%	6.004089e+06	1.000000	77.191964	28.570469	400.000000	2.000000	3.200000	31.000000
75%	1.835229e+07	1.000000	77.282006	28.642758	700.000000	2.000000	3.700000	131.000000
max	1.850065e+07	216.000000	174.832089	55.976980	800000.000000	4.000000	4.900000	10934.000000

```
# Fill missing cuisines with 'Unknown'
df['Cuisines'] = df['Cuisines'].fillna('Unknown')

# Check again
df.isnull().sum()
```

	0
Restaurant ID	0
Restaurant Name	0
Country Code	0
City	0
Address	0
Locality	0
Locality Verbose	0
Longitude	0
Latitude	0
Cuisines	0
Average Cost for two	0
Currency	0
Has Table booking	0
Has Online delivery	0
Is delivering now	0
Switch to order menu	0
Price range	0
Aggregate rating	0
Rating color	0
Rating text	0
Votes	0

dtype: int64

```
# Check duplicate rows
df.duplicated().sum()
```

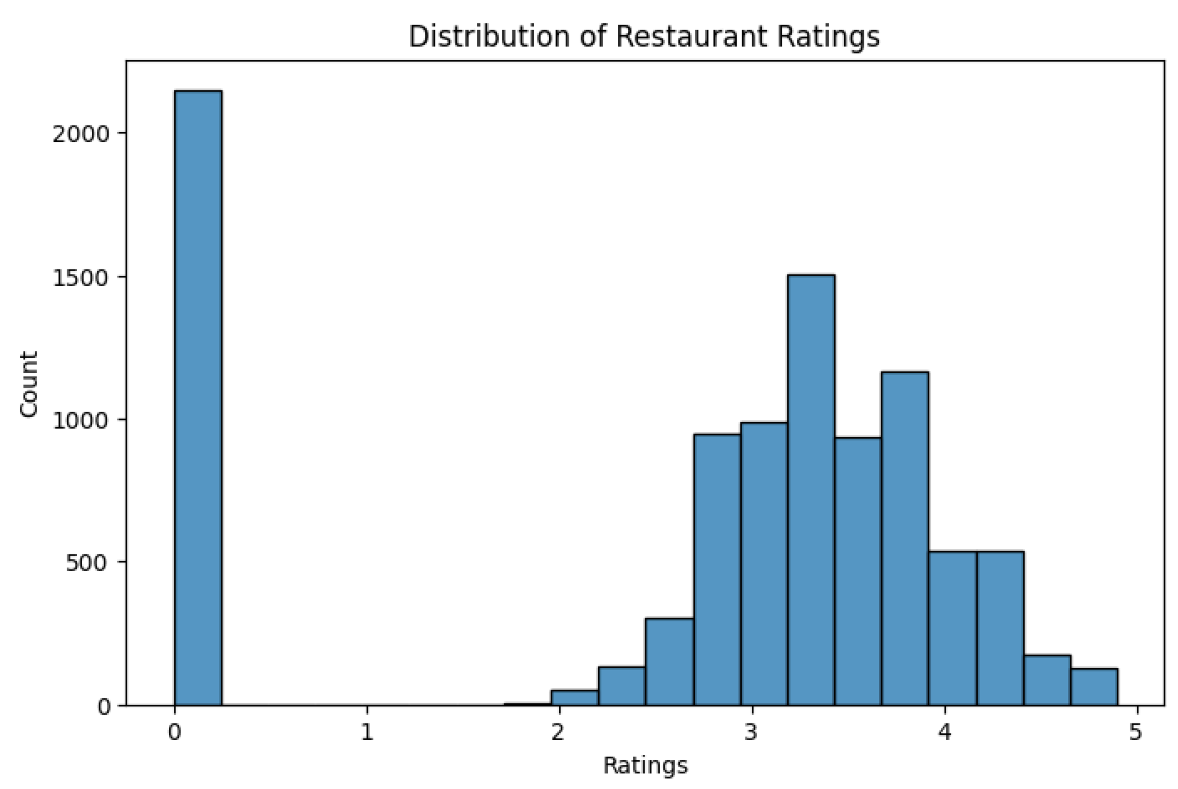
```
np.int64(0)
```

```
# Remove duplicates
df = df.drop_duplicates()
```

```
plt.figure(figsize=(8,5))
sns.histplot(df['Aggregate rating'], bins=20)

plt.title("Distribution of Restaurant Ratings")
plt.xlabel("Ratings")
plt.ylabel("Count")

plt.show()
```




```
top_cities = df['City'].value_counts().head(10)

print(top_cities)
```

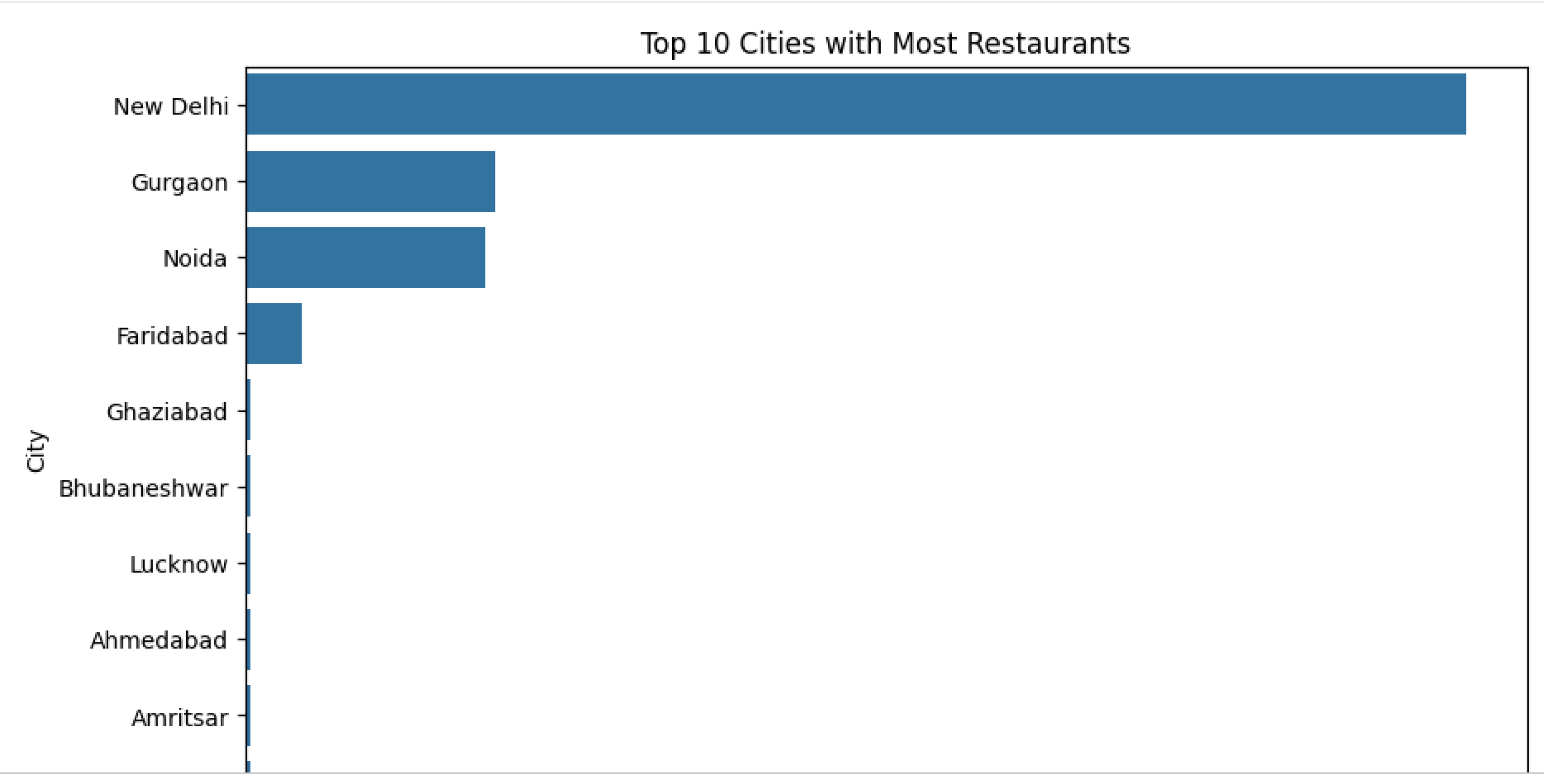
```
City
New Delhi       5473
Gurgaon         1118
Noida           1080
Faridabad        251
Ghaziabad         25
Bhubaneshwar      21
Lucknow           21
Ahmedabad         21
Amritsar          21
Guwahati          21
Name: count, dtype: int64
```

```
plt.figure(figsize=(10,6))

sns.barplot(
    x=top_cities.values,
    y=top_cities.index
)

plt.title("Top 10 Cities with Most Restaurants")
plt.xlabel("Number of Restaurants")
plt.ylabel("City")

plt.show()
```



```
city_ratings = df.groupby('City')['Aggregate rating'].mean().sort_values(ascending=False).head(10)

print(city_ratings)
```

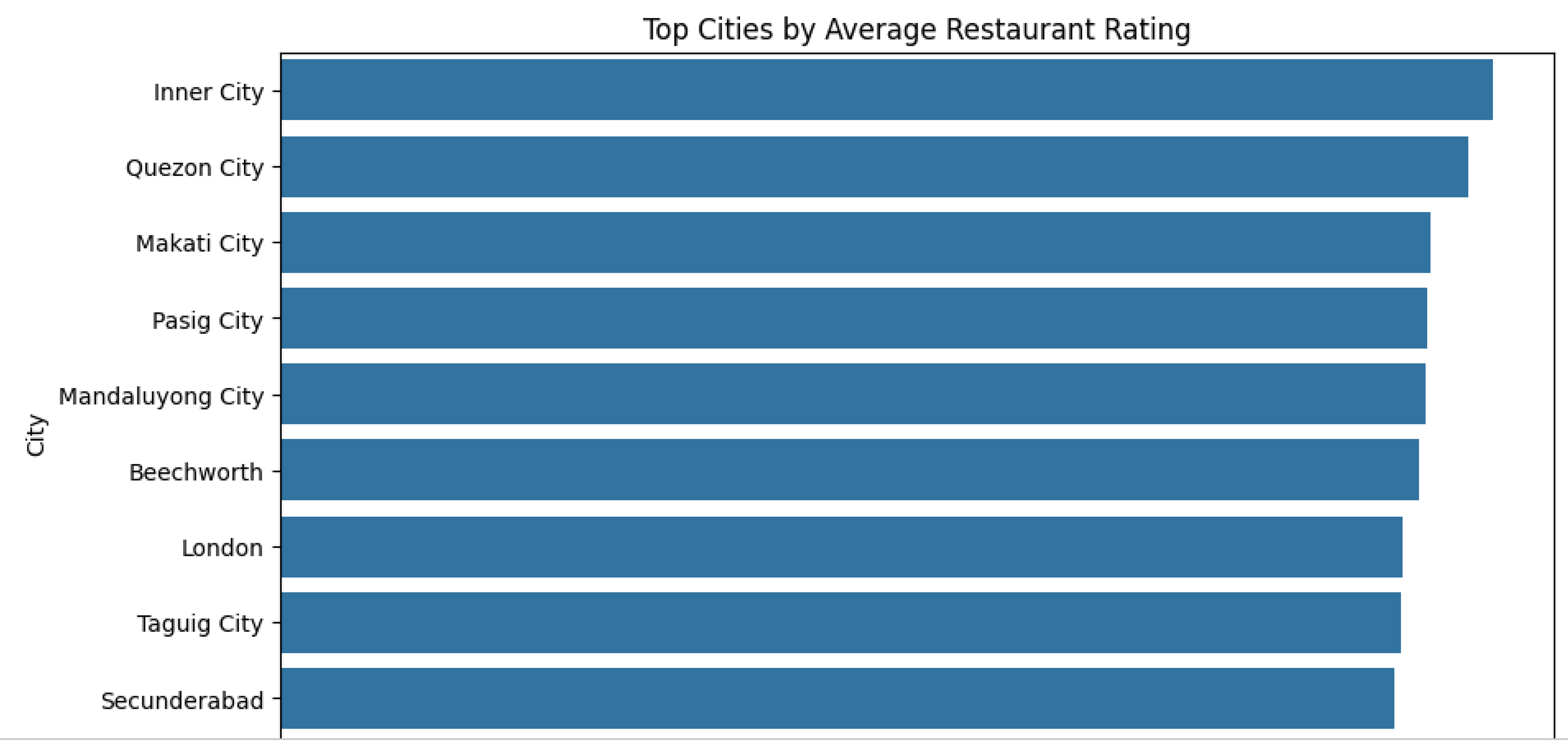
```
City
Inner City          4.900000
Quezon City         4.800000
Makati City         4.650000
Pasig City          4.633333
Mandaluyong City    4.625000
Beechworth          4.600000
London              4.535000
Taguig City         4.525000
Secunderabad        4.500000
Lincoln             4.500000
Name: Aggregate rating, dtype: float64
```

```
plt.figure(figsize=(10,6))

sns.barplot(
    x=city_ratings.values,
    y=city_ratings.index
)

plt.title("Top Cities by Average Restaurant Rating")
plt.xlabel("Average Rating")
plt.ylabel("City")

plt.show()
```




```
top_cuisines = df['Cuisines'].value_counts().head(10)

print(top_cuisines)
```

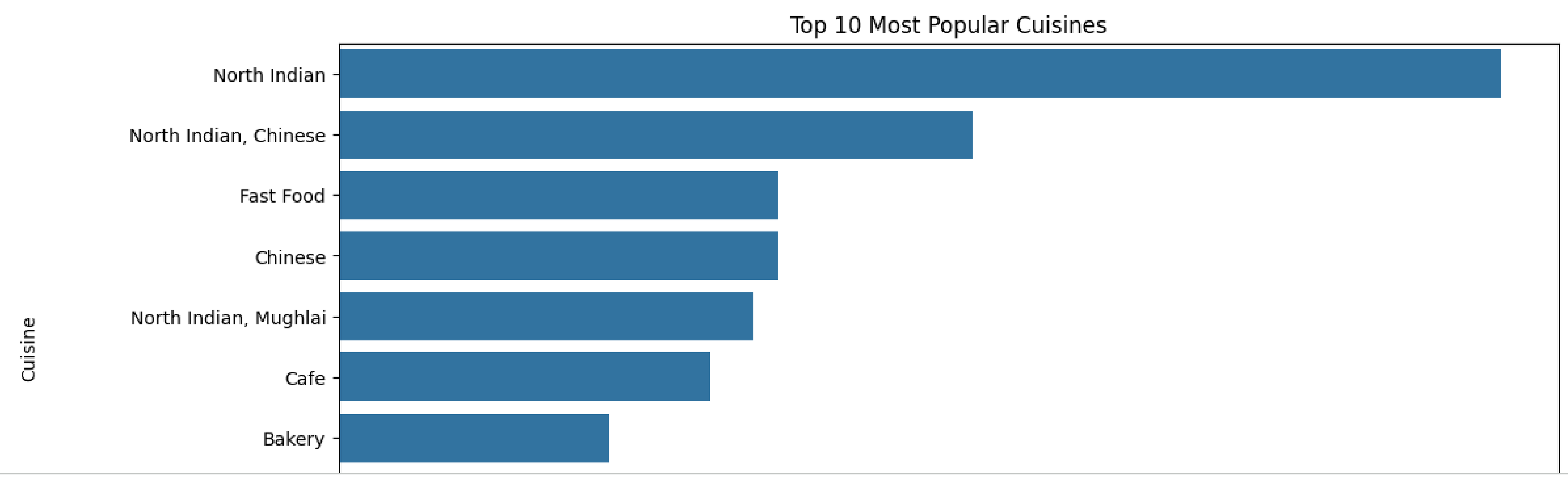
```
Cuisines
North Indian                      936
North Indian, Chinese             511
Fast Food                         354
Chinese                           354
North Indian, Mughlai             334
Cafe                              299
Bakery                            218
North Indian, Mughlai, Chinese    197
Bakery, Desserts                  170
Street Food                       149
Name: count, dtype: int64
```

```
plt.figure(figsize=(12,6))

sns.barplot(
    x=top_cuisines.values,
    y=top_cuisines.index
)

plt.title("Top 10 Most Popular Cuisines")
plt.xlabel("Count")
plt.ylabel("Cuisine")

plt.show()
```




```
plt.figure(figsize=(10,6))

plt.scatter(
    df['Longitude'],
    df['Latitude'],
    alpha=0.5
)

plt.title("Restaurant Locations")
plt.xlabel("Longitude")
plt.ylabel("Latitude")

plt.show()
```

Insights and Observations

1. Restaurant locations are concentrated in specific urban regions.
2. Certain cities contain significantly more restaurants than others.
3. Restaurant ratings are mostly clustered between average and high values.
4. Some cuisines appear much more frequently in the dataset.
5. Geographic clustering indicates higher restaurant density in developed city areas.
6. Location data can help businesses understand customer concentration and expansion opportunities.

Conclusion

This project performed a location-based analysis of restaurants using geographical and statistical methods.

The analysis explored:

- Restaurant distributions
- City-wise restaurant concentration
- Average ratings
- Popular cuisines
- Geographic patterns

Visualizations helped identify important trends and insights from the dataset.